

A disproof of Conjecture 00000007795

The unit square has Lipschitz boundary and lattice error $2m + 1$ at integer dilations m , not $O(m^{1/3})$

Jackmeson1

1 The conjecture

Conjecture 00000007795 reads (English text, verbatim): “Definition: Regularity stratification of lattice counting: the error $E(t) = |tP \cap \mathbb{Z}^n| - t^n \text{vol}(P)$ for the dilation tP . Conjecture: The optimal error for Lipschitz boundaries is $O(t^{n-2+1/3})$ and for C^2 boundaries $O(t^{n-2})$; no intermediate regularity exists between the exponents $1/3$ and 0 (any Holder exponent α in $(0, 1)$ has the error order jumping when α crosses $1/2$), the jump location being jointly determined by Fourier damping and the Diophantine properties of the lattice.” The Chinese text has the same content.

The statement is a conjunction. We refute its first conjunct, the *Lipschitz clause*:

(L) for every n and every body $P \subset \mathbb{R}^n$ with Lipschitz boundary, $E_P(t) = O(t^{n-2+1/3})$ as $t \rightarrow \infty$.

A false conjunct makes the conjunction false. Reading “the optimal error for the class is $O(\cdot)$ ” as a uniform bound over the class is stronger than the per-body bound (L), so it is refuted as well.

2 Definitions

We work in Euclidean space $\mathbb{R}^n$ (Lean: `EuclideanSpace R (Fin n)`).

Definition 1 (Lattice error; Lean: `latticeCount, discrepancy`). For $S \subseteq \mathbb{R}^n$, $|S \cap \mathbb{Z}^n|$ is the number of integer points in S , and $E_P(t) = |tP \cap \mathbb{Z}^n| - t^n \text{vol}(P)$ with $tP = \{tx : x \in P\}$ and vol the Lebesgue measure.

Definition 2 (Lipschitz boundary; Lean: `HasLipschitzBoundary`). A set $P \subseteq \mathbb{R}^n$ has Lipschitz boundary if for every boundary point x_0 there are $r > 0$, a unit vector v , $L \geq 0$ and an L -Lipschitz function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ that is constant along v ($\varphi(x + sv) = \varphi(x)$ for all x, s) such that

$$P \cap B(x_0, r) = \{x \in B(x_0, r) : \langle x, v \rangle \leq \varphi(x)\}.$$

A function constant along v is the same thing as a function of the coordinates in the hyperplane $v^\perp$, and it is L -Lipschitz on $\mathbb{R}^n$ exactly when it is L -Lipschitz on $v^\perp$. So the definition says: in an orthonormal frame whose last axis is v (rotations allowed), P is locally the region on one side of the graph of a Lipschitz function. This is the local Lipschitz-graph definition with ball neighbourhoods; P is closed, so the inequality is non-strict.

Remark 3 (The cylinder form). Wikipedia’s article “Lipschitz domain” (https://en.wikipedia.org/wiki/Lipschitz_domain) states the definition for a domain Ω with cylinder neighbourhoods: “for every point $p \in \partial\Omega$ there exists a hyperplane H of dimension $n - 1$ through p , a Lipschitz-continuous function $g : H \rightarrow \mathbb{R}$ over that hyperplane, and reals $r > 0$ and $h > 0$ such that”

$\Omega \cap C = \{x + y\vec{n} : x \in B_r(p) \cap H, -h < y < g(x)\}$ and $(\partial\Omega) \cap C = \{x + y\vec{n} : x \in B_r(p) \cap H, g(x) = y\}$, where $C = \{x + y\vec{n} : x \in B_r(p) \cap H, -h < y < h\}$. For our witness this form also holds, with $\Omega = (0, 1)^2$ the interior of the square: the chart φ used below is 1-Lipschitz along $v^\perp$ and satisfies $\varphi(p) = \langle p, v \rangle$ at boundary points, so with $r = 1/16$ and $h = 1/8$ the graph stays inside the cylinder, the cylinder lies in $B(p, 1/4)$ (as $\sqrt{r^2 + h^2} < 1/4$), and the strict and equality versions of the ball identity give the two cylinder identities. This remark is not formalized; the Lean development uses the ball form above.

3 The witness and the proof

Let $P = [0, 1]^2 \subset \mathbb{R}^2$ (Lean: `unitSquare`).

Lemma 4 (Lean: `isCompact_unitSquare`, `convex_unitSquare`, `interior_unitSquare_nonempty`). *P is compact and convex, and its interior is nonempty.*

Proof. P is the image of $[0, 1]^2$ under the continuous map to Euclidean space; convexity is coordinatewise; the ball of radius $1/2$ about $(1/2, 1/2)$ lies in P since $|x_i - y_i| \leq \|x - y\|$. $\square$

Lemma 5 (Lean: `corner_chart`, `coord_chart`, `unitSquare_hasLipschitzBoundary`). *P has Lipschitz boundary.*

Proof. Fix x_0 and $r = 1/4$. For the first coordinate put $(a, c_A) = (-1, 0)$ if $x_{0,1} \leq 1/2$ and $(a, c_A) = (1, 1)$ otherwise. On $B(x_0, 1/4)$ we have $|x_1 - x_{0,1}| < 1/4$, so the far constraint holds automatically and $0 \leq x_1 \leq 1 \iff ax_1 \leq c_A$. Define (b, c_B) from the second coordinate in the same way. Let $v = (a, b)/\sqrt{2}$ (a unit vector) and

$$\varphi(x) = (c_A + c_B - |ax_1 - bx_2 - c_A + c_B|)/\sqrt{2}.$$

With $X = ax_1 - c_A$, $Y = bx_2 - c_B$ one has $\sqrt{2}\langle x, v \rangle = X + Y + c_A + c_B$, so $\langle x, v \rangle \leq \varphi(x) \iff |X - Y| \leq -(X + Y) \iff X \leq 0$ and $Y \leq 0$. Hence $P \cap B = \{x \in B : \langle x, v \rangle \leq \varphi(x)\}$. Moving along v adds $s(a^2 - b^2)/\sqrt{2} = 0$ to $ax_1 - bx_2$, so φ is constant along v ; and $|\varphi(x) - \varphi(y)| \leq |x_1 - y_1| + |x_2 - y_2| \leq 2\|x - y\|$, so φ is 2-Lipschitz. $\square$

Lemma 6 (Lean: `volume_unitSquare`, `latticeCount_unitSquare`, `discrepancy_unitSquare`). $\text{vol}(P) = 1$, and for every integer $m \geq 1$, $|mP \cap \mathbb{Z}^2| = (m + 1)^2$, hence $E_P(m) = 2m + 1$.

Proof. The map from Euclidean space to $\mathbb{R}^2$ with the product measure preserves volume, and $\text{vol}([0, 1]^2) = 1$. A lattice point z lies in mP iff $0 \leq z_i/m \leq 1$, i.e. $0 \leq z_i \leq m$, for $i = 1, 2$; there are $(m + 1)^2$ such z . So $E_P(m) = (m + 1)^2 - m^2 = 2m + 1$. $\square$

Theorem 7 (Lean: `not_isBig0_nat`, `conjecture7795_false`). $E_P(m) = 2m + 1$ is not $O(m^{1/3})$ as $m \rightarrow \infty$ through the integers. Consequently (L) fails for $n = 2$ (where $n - 2 + 1/3 = 1/3$), both for real $t \rightarrow \infty$ and for integer dilations.

Proof. Suppose $2m + 1 \leq Cm^{1/3}$ for all $m \geq N$. Take an integer $m \geq N$ with $m \geq 1$ and $m > C^3$ and put $u = m^{1/3}$, so $u \geq 1$ and $u^3 = m$. From $u^3 = m > C^3$ and $u \geq 0$ we get $u > C$, so $2u^3 + 1 \leq Cu < u^2 \leq u^3$, a contradiction. A bound for real $t \rightarrow \infty$ would restrict to integer $t = m$ (composition with $m \mapsto m$, which tends to infinity), so it fails too. $\square$

4 Lean formalization

File `lean/Conjecture7795/Basic.lean` (Lean 4, Mathlib v4.33.1, only import `Mathlib`, 297 lines). The clause (L) is formalized as `LipschitzClause` (real $t \rightarrow \infty$) and `LipschitzClauseNat` (integer $m \rightarrow \infty$): for all n and all compact $P \subseteq \mathbb{R}^n$ with nonempty interior and `HasLipschitzBoundary` P , $E_P = O(t^{n-2+1/3})$ along `atTop`. Main theorem (ASCII rendering):

```
theorem conjecture7795_false :
  IsCompact unitSquare /\ Convex R unitSquare /\
    (interior unitSquare).Nonempty /\ HasLipschitzBoundary unitSquare /\
    (forall m : N, 0 < m -> discrepancy unitSquare m = 2 * m + 1) /\
    Not LipschitzClauseNat /\ Not LipschitzClause
```

Axiom **audit.** `#print axioms C7795.conjecture7795_false` prints `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

Conventions. $|S \cap \mathbb{Z}^n|$ is `Set.ncard` (finite here); `vol` is Mathlib’s `volume` on Euclidean space, converted to a real number; $t^{n-2+1/3}$ is the real power `Real.rpow`; $O(\cdot)$ is Mathlib’s `IsBigO` along `atTop`.

Scope. Refuted: the Lipschitz clause (per body and hence uniform over the class), for real and integer dilations, in dimension $n = 2$. Not addressed: the C^2 clause and the Holder/“jump” clauses (not needed, since the conjunction already fails). The cube $[0, 1]^n$ gives $E(m) = (m + 1)^n - m^n \geq nm^{n-1}$ in every dimension, but only $n = 2$ is formalized.